Is photoperiod the dominant or least important cue for woody plant spring phenology?

Lizzie & Dan so far

November 10, 2026

Abstract

Photoperiod is critical to plant function, entraining the circadian clock, settling the length of sunlight for photosynthesis and other stuff. But how much it matters to woody plant leafout and flowering has become less clear given more and more data. That's a bummer....

Daylength is a critical controller of growth and reproduction in plants. It defines how long they can photosynthesize each day and determines flowering success in many species. While decades of research have identified the photoperiods different species appear to need for major events, such as leafout and flowering, new research suggests gaps in our understanding of exactly how photoperiod controls fundamental plant biology CITES. At the same time, anthropogenic climate change has shifted plant events across the globe as temperatures rise—driving a growing literature of when and how photoperiod may slow this acceleration CITES.

Recent climate change has challenged our biological understanding of the dual roles of temperature and daylength to plant phenology—the timing of recurring life history events, such as leafout and flowering—with major implications for forecasting. Because forests currently sequester a large portion of fossil fuel emissions, how much they shift their growth and how successfully they can reproduce with continued warming is critical to forecasting future climate change itself CITES. Further, accurate phenological forecasts guide many forest restoration programs and crop plantings, especially of woody plants CITES. If plants mainly respond to temperature then they are more likely to shifts in step with continued warming. In contrast, if they rely on photoperiod they are more likely to lag in their responses CITES. Teasing out how much woody plants rely on photoperiod versus temperature has thus become an increasingly important topic.

Decades of work before significant anthropogenic climate change identified three major cues that plants rely on to time leafout and flowering: chilling (cool winter temperatures), forcing (warm spring temperatures) and photoperiod. Studies of model systems in annual/biennial plants show how successful flowering generally requires vernalization (chilling that leads to flowering) to make sure flowering takes place at the correct time seasonally alongside an internal circadian clock that relies on photoperiod to further determine timing CITES. These required—or 'critical photoperiods'—give rise to the terms 'short-day' and 'long-day' plants. Plants grown without vernalization or required daylengths will often flower at less optimal times climatically and/or fail to flower altogether CITES (AND reskim Wilcek). Forcing effectively drives higher rates of cellular processes

and photosynthesis, producing flowing in the end, after timing determined by vernalization and circadian clock pathways CITES.

This general model is widely assumed to operate for woody plant leafout and flowering as well CITES, but has been studied far less closely. While studies in annual/biennial plants have designed experiments to separate out effects of an actual photoperiodic trigger from effects of increased photosynthesis (e.g., though night-interruption and manipulating the type and quality of light), experiments with woody plants almost never apply such methods CITES(BUT see Okie, Japanese studies). Instead, they generally focus on factorial experiments that manipulate temperature and light, generally including one shorter (e.g., 8 hour) and longer (e.g., 16 hour) photoperiod treatments with continuous high light CITES(Ailene). This approach has the benefit of directly comparing the effect of temperature versus daylength, but cannot tease out the photosynthetic effects of photoperiod that may effective drive the event to occur (similar to forcing), from the presence of 'critical photoperiods' that trigger the event.

Whether woody plants have critical photoperiods that determine leafout and flowering has major implications for forecasts in agriculture, forestry and even for climate change itself, but is still widely debated. For over two decades studies have declared that photoperiod is either the most important CITES or least important cue for woody plant spring phenology. A recent meta-analysis of experiments on woody plant leafout found the effect of photoperiod was far smaller than effects of chilling or forcing CITES, seeming to conclude the debate. Yet the study also found a consistent effect of photoperiod that appeared highly similar across most species.

The evolutionary signal in photoperiod hints at a conserved molecular basis for woody plant daylength responses, but one that mechanistically does not easily fit well within current photoperiod models. Current studies generally observe high leafout (or flowering) with both short and long photoperiods CITES, which does not fit with traditional circadian clock models (e.g., the internal coincidence model, CITES). Further, it seems unlikely to be solely a photosynthetic effect given how consistent it is across diverse species. It may thus potentially be derived from some other model of photoperiod response. Indeed, alongside this growing search in woody plant photoperiod responses, molecular biology has found annual/biennial plant responses that suggest missing models of photoperiod responses.

Here we compare current results on the role of daylength in woody plants with models of photoperiod from plant biology to identify where they do—or do not—seem to fit. To do this, we first briefly review well established models of photoperiod control and how they are often invoked in woody plant phenology research. Then we review new models from molecular biology that may match better to woody plant responses and discuss how a new generation of experiments could test across multiple models. Such experiments could quickly expand our understanding of woody plant phenology with major implications for forecasting.

Current well-established models of photoperiod control in plant leafout and flowering

Much work on daylength responses has focused on what controls the apparent critical photoperiod some plants need to flower, with photoperiods below this length leading to little flowering at all. Early work hypothesized an hourglass timer, where a hypothetical substance acts as the sand in an hourglass to measure either the night or day. This model predicts a certain number of light (or more often dark) hours should trigger normal flowering, even if the night is interrupted by a short period of light. But a series of night-interruption experiments showed this was clearly not the case and the hourglass model fell generally out of favor as circadian clock controlled timers appeared more common in plants, including the well-studied *Arabidopsis thaliana*.

Today, most daylength-controlled flowering appears to rely on a circadian clock timer, where a series of genes promote or suppress protein expression over a 24-hour period. Flowering is then triggered when the clock is in a certain phase and light is sensed (external coincidence model), or when the phases of two (or more) circadian clocks line up (internal coincidence models, seen more often in animals, CITEIMAIZUMI). In both cases light helps entrain the clocks. These models predict night-interruption experiments can disrupt flowering depending on the critical photoperiod of the plant (e.g., short-day plants would do WHAT under some night-break experiment). The external coincidence model appears to control flowering in many herbaceous plant species, with work over many decades categorizing species (and cultivars) into whether they require short or long photoperiods to flower (short-day and lond-day, respectively), or appear unresponsive (day-neutral).

Recently, a new model of photoperiod response has been discovered that relies not on the traditional understanding of photoperiod, but on the shorter photosynthetic period. The photosynthetic period represents when photosynthesis is possible (light is above the compensation point, cite LIANDGENDRON2024). Termed the 'Metabolic Daylength Measurement System,' this pathway appears related to rapid vegetative growth under long days and suggests plants measure multiple photoperiods, potentially to time different processes. It also suggests a potentially larger role for plant metabolism (specifically starch and sugar regulation) in how photoperiod controls plant growth, with potential links to increasing evidence that plant metabolism may affect dormancy and frost hardiness CITES.

Does photoperiod affect woody plant spring phenology?

Plant biology has long recognized the importance of photoperiod in triggering critical plant events, but anthropogenic climate change has challenged ...

Leafout and flowering are some of the most important events for plants, with their timing determining the success of growth and reproduction. In the temperate zone, these events often happen in the spring, when climate poses potential risks via frost but early-activity can provide priority access to resources, including sunlight and nitrogen. Spring is also a time when humans like to monitor stuff...

References

Figures

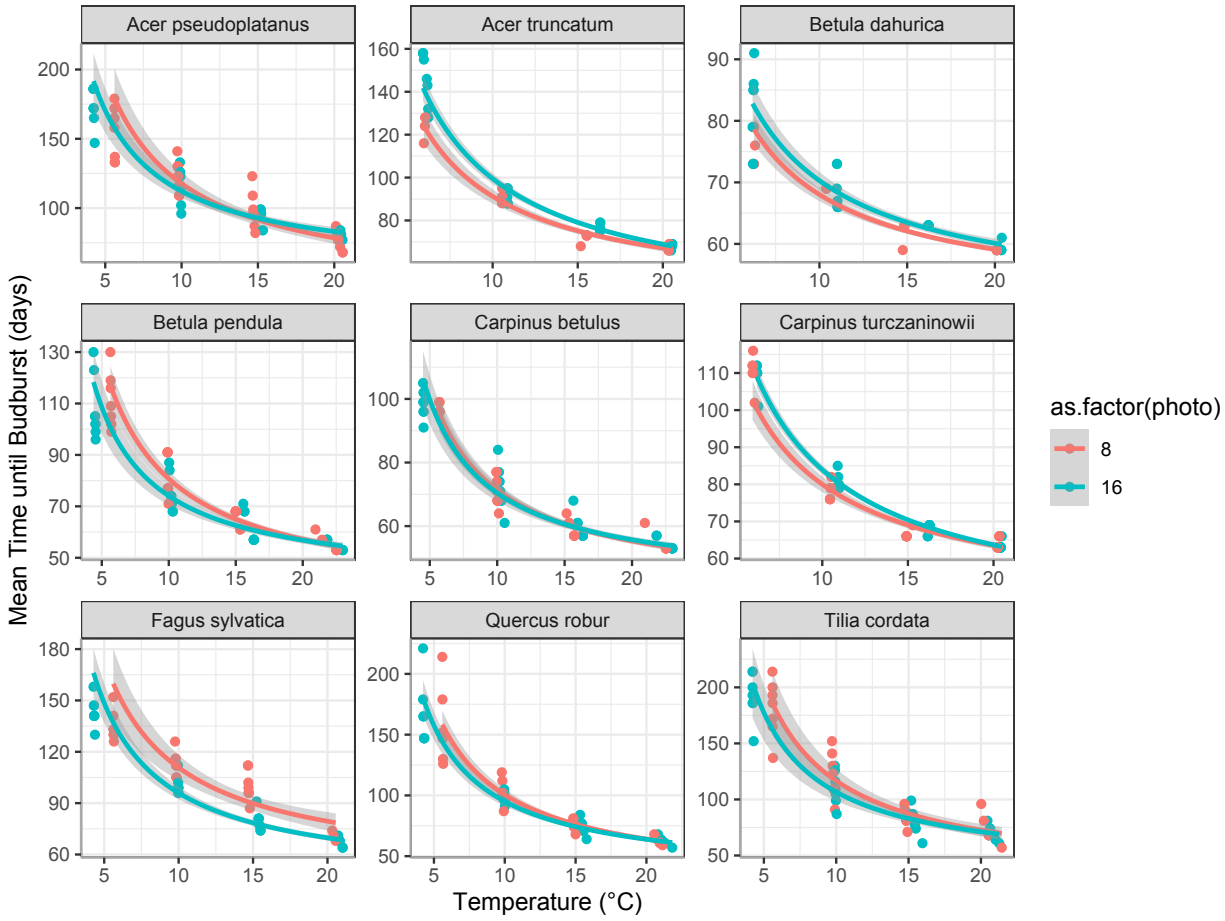


Figure 1: Results from Walde *et al.* 2023 for budburst timing across species under high chilling and two photoperiods: 8 and 16 hours. This type of study is the most common type done in woody plant phenology—it uses cuttings of dormant buds exposed to chilling (cool temperatures) followed by different combinations of warm temperatures and photoperiods (this study is unique though in covering a wide range of forcing temperatures).

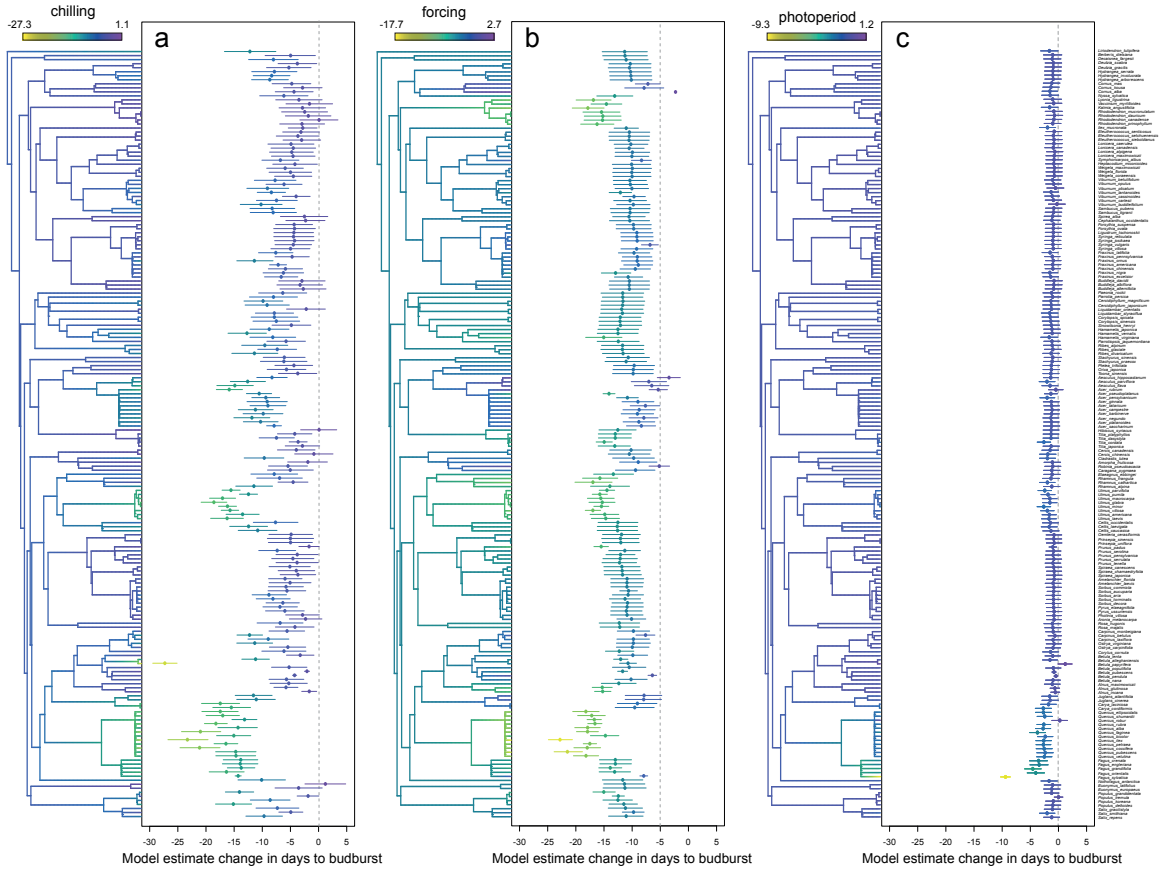


Figure 2: Estimated chilling, forcing and photoperiod responses (in units standardized by the mean and SD for each response variable) from 191 woody angiosperm species show most species shift earlier two days per standardized photoperiod unit, except for *Fagus* species with the well-studied *Fagus sylvatica* showing the strongest response. Figure reproduced from **?**.